

Independent Lab Work No. 1

Web Scraping & Descriptive Data Analysis

Due date: 19.02.2026

Submission:

- Report (PDF)
- Dataset (CSV/JSON)
- Code (Notebook or Script)

1. Data Collection: Web scraping.
 - Choose any website from the list of ethical scraping websites in Table 1 below that allows web scraping.
 - Use any web scraping tool (e.g., BeautifulSoup, Scrapy, Selenium, or any other) to extract meaningful data.
 - Save the extracted data in a structured format such as CSV or JSON.
2. Your dataset must have at least 500 rows and multiple features (columns).

Suggested Websites for Web Scraping

- <https://books.toscrape.com/>
 - <https://quotes.toscrape.com/>
3. Using descriptive statistics methods on the extracted data:
 - Summarize the dataset (including statistics as: mean, median, mode, standard deviation, etc.)
 - Identify missing values and handle them using either deletion or imputation.
 - Create at least three graphical or visualizations of the dataset:
 - Box plot to show the distribution of numerical data.
 - Bar chart to display categorical variable distribution.
 - Scatter plot to analyze relationships between two numerical variables.

S/N	Website	Description of website
1	https://books.toscrape.com/	online bookstore with book titles, prices, ratings
2	https://quotes.toscrape.com/	text-based website with quotes and authors
3	https://world.openfoodfacts.org/	open-source food product database
4	https://www.kaggle.com/datasets	open datasets repository